

# Frequency-Response – Worked Problems

**Practice:** read margins, find the max gain for a spec, and design a lead.

Ogata, *Modern Control Engineering*, Ch. 7 (end-of-chapter style).

Step through with **Ctrl+Enter**, or render a report with **publish**.

## Contents

- Problem 1: Find Gain/Phase Margins and Assess Stability
- Problem 2: Maximum Gain for a Phase-Margin Specification
- Problem 3: Lead Compensator to Meet a Phase-Margin Spec
- Try it yourself

### Problem 1: Find Gain/Phase Margins and Assess Stability

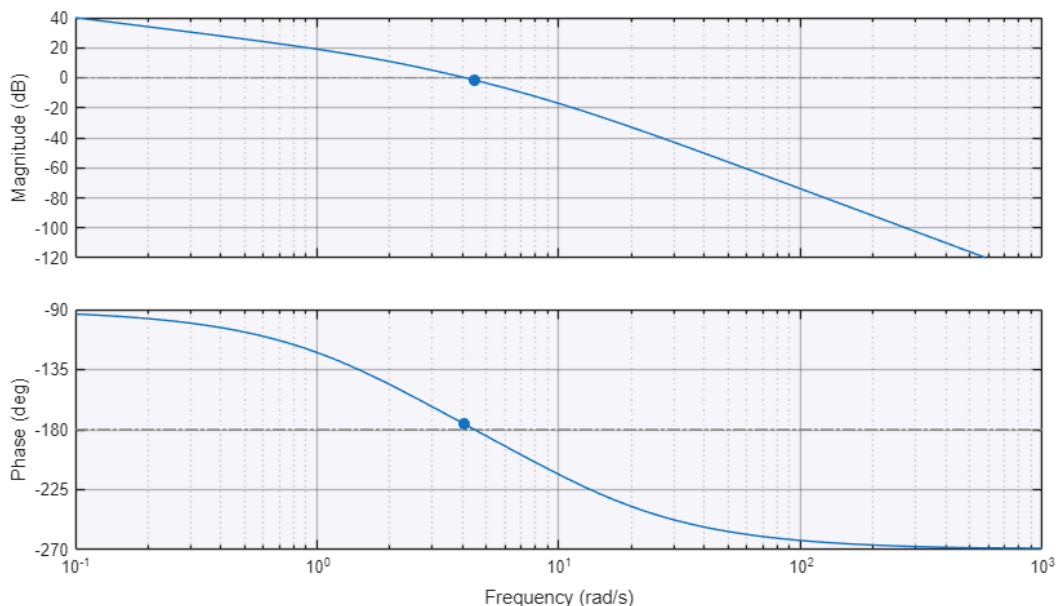
For  $G(s) = \frac{K}{s(s+2)(s+10)}$  with  $K = 200$ , find the gain and phase margins and determine whether the closed-loop system is stable.

```
K1 = 200;
G1 = tf(K1,conv([1 0],conv([1 2],[1 10])));
[Gm1,Pm1,Wcp1,Wcg1] = margin(G1);
fprintf('Problem 1: GM = %.4f (%.2f dB) at wp=%.4f, PM = %.2f deg at wc=%.4f\n', ...
        Gm1, 20*log10(Gm1), Wcp1, Pm1, Wcg1)
fprintf('Stable (both margins positive): %d\n', (Gm1>1) && (Pm1>0))
```

```
figure
margin(G1)
title('Problem 1: Gain/Phase Margins','Interpreter','latex','FontSize',20)
grid on
```

```
Problem 1: GM = 1.2000 (1.58 dB) at wp=4.4721, PM = 3.94 deg at wc=4.0776
Stable (both margins positive): 1
```

**Problem 1: Gain/Phase Margins**  
Gm = 1.58 dB (at 4.47 rad/s), Pm = 3.94 deg (at 4.08 rad/s)

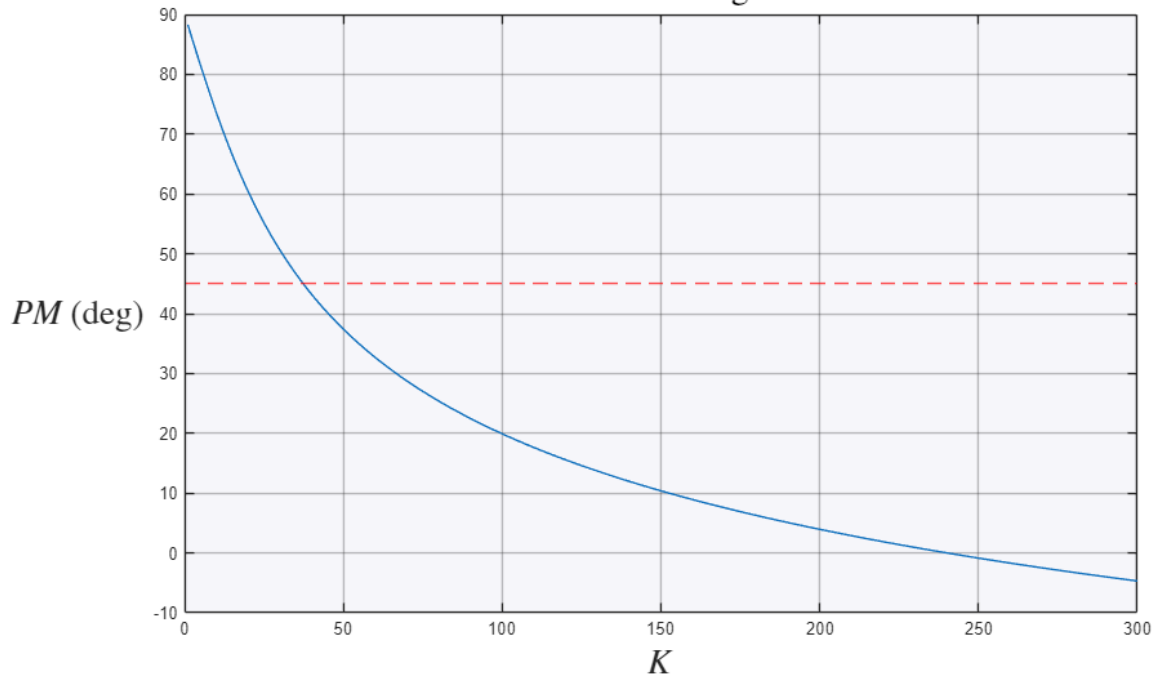


For the same plant family  $G(s) = \frac{K}{s(s+2)(s+10)}$ , find the largest  $K$  for which  $PM \geq 45^\circ$ .

```
figure
plot(K_range,PMs)
hold on
yline(45,'r--')
hold off
title('Problem 2: Phase Margin vs. Gain','Interpreter','latex','FontSize',20)
ylabel('$PM$ (deg)','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$K$', 'Interpreter','latex','FontSize',20)
grid on
```

2

## Problem 2: Phase Margin vs. Gain



## Problem 3: Lead Compensator to Meet a Phase-Margin Spec

For  $G(s) = \frac{10}{s(s+1)}$ , design a lead compensator so that the phase margin is at least  $40^\circ$  while preserving the existing velocity-error constant  $K_v = 10$ .

```
G3 = tf(10,[1 1 0]);
PM_des3 = 40;
[~,PM1_3,~,wc1_3] = margin(G3);
fprintf('Problem 3: Uncompensated PM = %.2f deg at wc=%.2f\n', PM1_3, wc1_3)

epsilon3 = 10;
phi_max3 = (PM_des3 - PM1_3 + epsilon3)*pi/180;
alpha3 = (1+sin(phi_max3))/(1-sin(phi_max3));

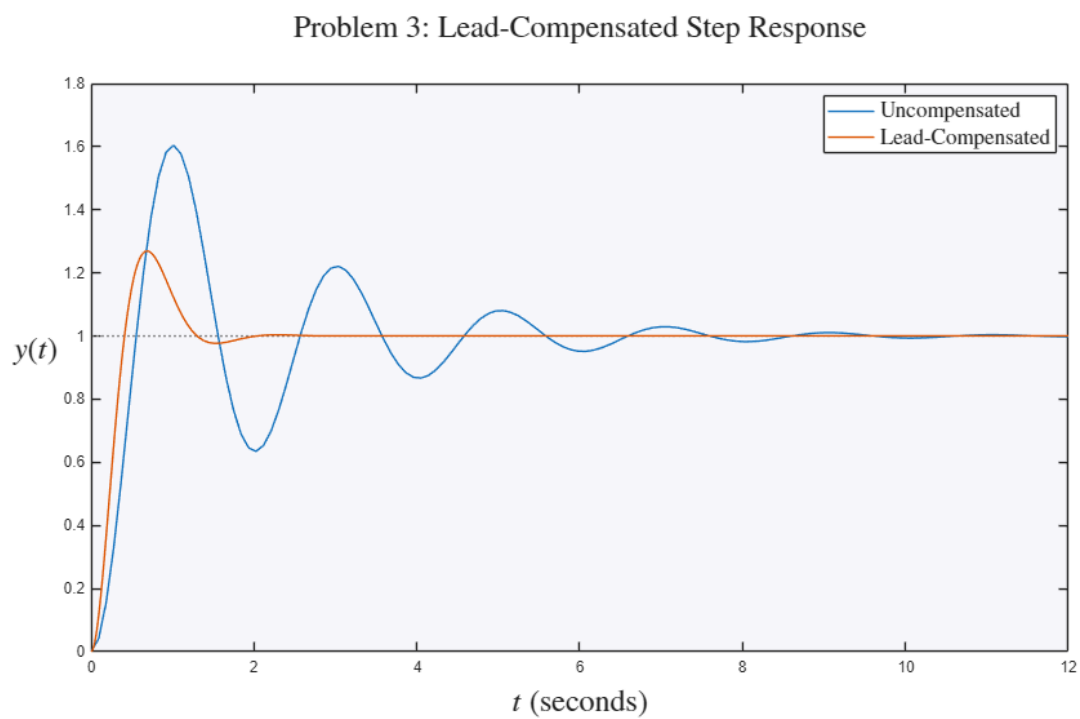
w_sweep3 = logspace(-1,2,2000);
[mag3,~] = bode(G3,w_sweep3);
mag_dB3 = 20*log10(squeeze(mag3));
mag_needed_dB3 = -10*log10(alpha3);
[~,idx3] = min(abs(mag_dB3 - mag_needed_dB3));
wm3 = w_sweep3(idx3);
T3 = 1/(wm3*sqrt(alpha3));

Gc3 = tf([alpha3*T3 1],[T3 1]);
G3_comp = series(Gc3,G3);
[~,PM3_comp,~,wc3_comp] = margin(G3_comp);
fprintf('Problem 3: alpha=%.4f, T=%.4f, Compensated PM = %.2f deg at wc=%.2f\n', ...
        alpha3, T3, PM3_comp, wc3_comp)

T3_uncomp = feedback(G3,1);
T3_comp_cl = feedback(G3_comp,1);
figure
hold on
step(T3_uncomp)
step(T3_comp_cl)
hold off
legend('Uncompensated','Lead-Compensated','Interpreter','latex','FontSize',14)
title('Problem 3: Lead-Compensated Step Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)
```

Problem 3: Uncompensated PM = 17.96 deg at  $\omega_c=3.08$

Problem 3:  $\alpha=3.2594$ ,  $T=0.1322$ , Compensated PM = 45.46 deg at  $\omega_c=4.19$



### Try it yourself

- In Problem 1, halve  $K$  and confirm both margins grow (a more sluggish, more robust loop).
- In Problem 2, tighten the spec to  $\text{PM} \geq 55$  and read off the new maximum allowable gain from the sweep.